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Generative AI chatbot for Student Counselling

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Abstract

Mental illnesses now impact people in all nations and have become a widespread global problem. However, a major obstacle to resolving this issue is the severe lack of personnel committed to providing vital mental health care. Due to this shortage, a sizable fraction of people with mental illnesses goes untreated, which has serious repercussions such as increased rates of suicide attempts and mortality. The use of conversational agents has significantly increased recently because of the problem of scarce resources. These artificial intelligence (AI)-driven systems, such as chatbots and virtual mental health counsellors, provide scalable and accessible methods of providing mental health care. For people in need, particularly in rural areas, they can offer advice, direction, and even therapeutic interventions, making mental health support more accessible.

Introduction

In our increasingly interconnected world, the global prevalence of mental illnesses presents a formidable challenge – a severe shortage of mental health care providers. This deficiency has far-reaching consequences, with many individuals left untreated, leading to elevated suicide rates and increased mortality. To address this crisis, the deployment of AI-driven conversational agents, such as chatbots and virtual mental health counselors, has surged. These systems offer scalable and easily accessible methods for delivering crucial mental health support, particularly benefiting those in remote or underserved areas.

Motivation

Mental illnesses are pervasive and transcending borders, impacting people from all walks of life. However, the shortage of mental health professionals presents a daunting obstacle, leaving a substantial portion of the population without access to vital care. This deficiency has severe consequences, including elevated rates of suicide attempts and increased mortality. In response to this urgent challenge, our project seeks to leverage AI-driven conversational agents, such as chatbots and virtual mental health counselors, to provide scalable and easily accessible mental health support. By doing so, we aim to bridge the gap, ensuring that individuals, particularly those in remote or underserved areas, receive the critical assistance they need to lead healthier and happier lives.

Literature Review

PAPER TITLE	OBJECTIVE	METHODOLOGY	ADVANTAGES	LIMITATIONS/ FUTURE SCOPE
A Mental Health Chatbot for Regulating Emotions (SERMO) - Concept and Usability Test 1 July 2021 IEEE	Address the global shortage of human resources delivering mental health services by introducing a mobile application called SERMO	SERMO to provide CBT-based emotional support via NLP and daily interactions.	Personalized, continuous monitoring, education, and self-help for global mental health support	Lexicon-based approach for emotion detection in NLP may not accurately capture complex or subtle emotions
AI-Based Deep Learning Chatbot for Career and Personal Mentorship 16 June 2023 IEEE	virtual mentor chatbot for career guidance and emotional support	Creating a dataset for training and utilizing TensorFlow and Keras for machine learning models. HTML and CSS are used for the graphical user interface, while Flask integrates the system into a user-friendly web application	This system serves as a convenient and accessible alternative to traditional mentoring, providing students with tailored support	While it offers anonymity and convenience, it may lack the personalized, empathetic touch of in-person mentoring sessions.
Machine Learning based Intelligent Career Counselling Chatbot (ICCC). 24 May 2023	Intelligent Career Counselling Chatbot (ICCC) capable of providing career guidance to 10th and 12th-grade students. Additionally, the chatbot aims to assist students in making informed	user-centric design, data selection from YML files, and the utilization of the Chatterbot module for conversational interaction. Machine learning algorithms, including Decision Tree, Random	The utilization of the Random Forest algorithm increases prediction reliability by aggregating output from multiple trees	While machine learning algorithms improve accuracy, they can also introduce complexity, potentially

Career Counseling Chatbot using Microsoft Bot Frameworks1 11 May 2022 IEEE	Aims to support students in identifying and planning their future career paths based on their profiles	Microsoft technologies, including QnA , Bot Composer for integration	QnA Maker allows easy creation of a knowledge base for tailored responses.	Licensing and operational costs for Azure services may be involved, depending on usage and scale.
Mind Relaxation Chatbot for University Students by Using Dense Neural Network 5 January 2022 IEEE	Aims to address this issue by creating a chatbot that can provide a non-judgmental and supportive platform for students to express their thoughts and worries openly	Data collection, preprocessing, DNN model for chatbot development using NLP	Give suggestions to students to overcome stress during academic pressure and deadline on projects, helping them to manage user experience.	Ethical considerations arise when using chatbots in sensitive areas like mental healthcare.
AI-Driven Bilingual Talkbot for Academic Counselling 24 July 2023 IEEE	Develop a multilingual Talbot for efficient, domain-specific, and responsive user support.	deep learning, NLP, and domain-specific knowledge to create a user-friendly, multilingual interface.	The use of a domain-specific knowledge base allows for tailored responses to specific educational queries.	The use of translation APIs for language conversion, especially for Arabic, may introduce errors or inconsistencies in the responses.
Dost-Mental Health Assistant Chatbot 13 February 2023 IEEE	To provide accessible mental health support and guidance through empathetic and informative conversations.	Employs Rasa's NLU pipeline, leveraging AI, natural language processing, and emotion recognition to understand and respond	The chatbot offers easily accessible mental health support to a wide range of users via the Telegram application, the chatbot's ability to use predefined stories and rules allows for a tailored and flexible user experience.	The chatbot may have limitations in addressing complex mental health issues, as it primarily relies on predefined responses and rules

Existing Challenges

From a research point of view, the challenges faced in existing counselling chatbots include:

1. **Generalization:** Getting a general answer may not be apt for every user, instead it should be modifiable with respect to user prompt rather than just matching the words in data and giving answer.
2. **Maintaining History:** Not able to maintain a record or history of the previous conversation, which could lead to wrong/irrelevant results for a user using that chatbot.
3. **Hallucination:** Using LLM(large language models) can be helpful in giving human like answers , but it run into hallucination when not trained on validate data and keeping it up to date.
4. **Gratitude Journal:** Giving details of daily task to the chatbot is a difficult task for the users who have a busy schedule.

Objectives

The objectives of our project can be stated as:

- ❖ A chatbot which resolves the query of students at any given time, reducing the dependency/workload on human counsellor.
- ❖ Able to give the answer in simplest and easy to understandable form; mimicking humans.
- ❖ Generative that is will be able to give the answer to the query in multiple ways like tabular, different languages , points based on user prompt; also give the reference to the specific part of the documentation from which the answer has been extracted (reference).

Innovation Idea

We employ the formidable power of Azure OpenAI services, harnessing the remarkable capabilities of the GPT-3.5 model to deliver empathetic and contextually relevant responses to students seeking guidance. Azure Cognitive Search empowers our chatbot with the ability to index and search through a wealth of data swiftly, ensuring that the information provided is not just accurate but also comprehensive.

One distinctive aspect of our project is the use of Azure Blob Storage to house the data on which our model is trained, offering scalability, security, and easy access. Additionally, our chatbot goes a step further in enriching the user experience by returning relevant resources, data, and paragraphs from the underlying data sources. This unique feature enhances the effectiveness of the counseling, as students not only receive answers but also gain insights and access to valuable resources to support their mental well-being.



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Scope and application of the Project

❖ Scalability and Integration

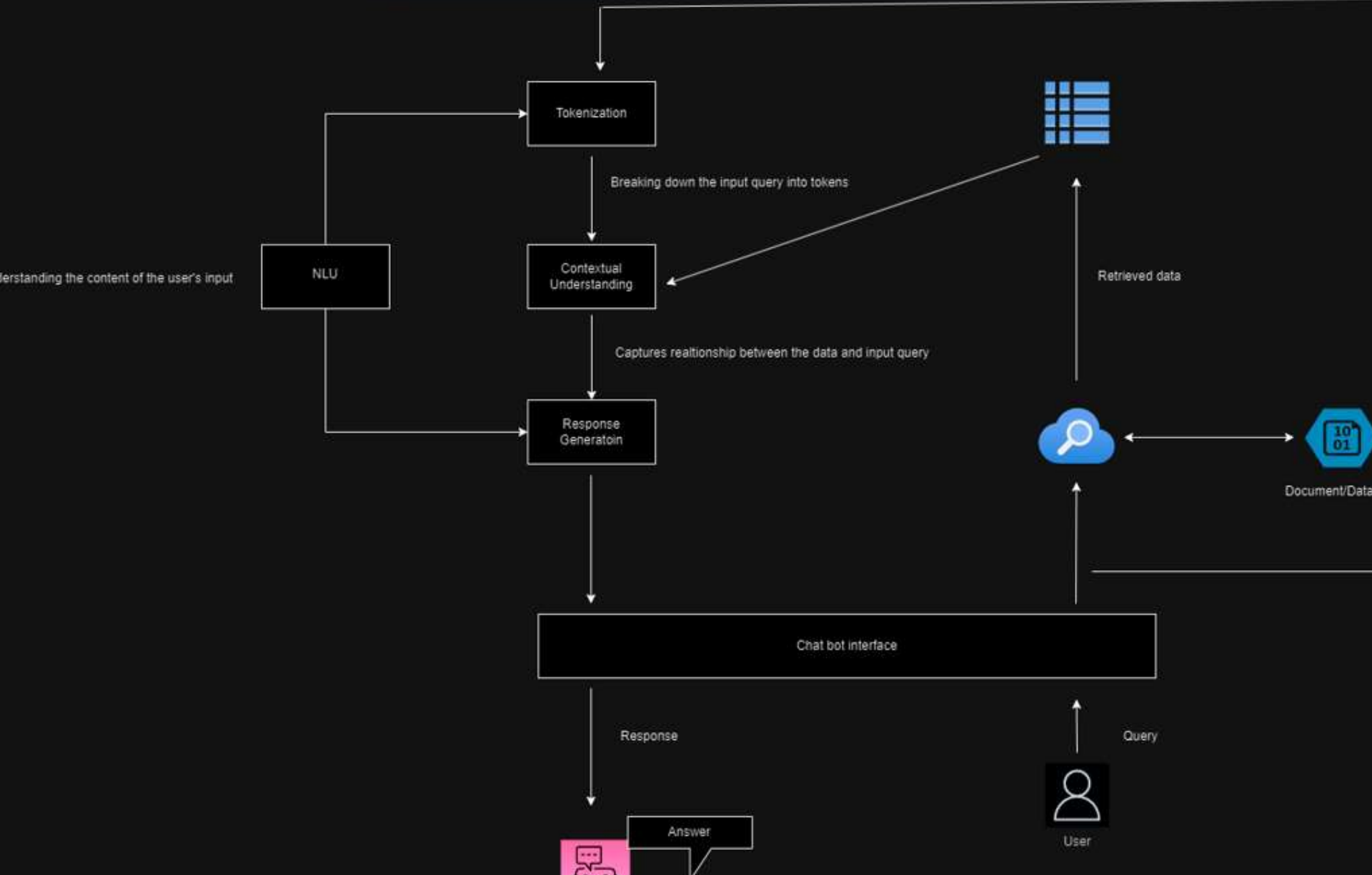
The technology's cloud-based architecture ensures scalability and adaptability, making it suitable for integration into various healthcare and educational systems. This project not only enhances the reach and quality of mental health support but also showcases the versatility and scalability of cloud-based solutions in addressing critical societal challenges.

❖ Valid data sources

With the ability to provide fast and accurate responses, drawing from extensive data sources, it's an invaluable tool for addressing mental health concerns among students.



Architecture



Algorithm

The underlying algorithm in GPT-3 (Generative Pre-trained Transformer 3) is a **deep neural network architecture** called a transformer.

The key innovation of the transformer architecture is the attention mechanism, which allows the model to focus on different parts of the input text when generating output. This mechanism greatly improved the performance of machine learning models in tasks such as machine translation, text generation, and language understanding

Modules

- ❖ **Input Preprocessing and Tokenization:** The initial text queries undergo preprocessing and tokenization, which breaks them down into manageable units. Able to give the answer in simplest and easy to understandable form; mimicking humans.
- ❖ **Contextual Comprehension via Transformer Architecture:** A GPT (Generative Pre-trained Transformer) model, like GPT-3, leverages a deep neural network based on the Transformer architecture. This architecture captures contextual nuances and relationships within the input text.
- ❖ **Natural Language Understanding (NLU):** The NLP module comprises a range of methods, including intent recognition, sentiment analysis, and entity recognition. These techniques are employed to comprehend and categorize student queries effectively.
- ❖ **Context Management:** Within the NLP module, this component is responsible for preserving a continuous conversation context, ensuring that the chatbot maintains a coherent understanding and delivers relevant responses.

Working of Transformer

- ❖ The input sequence is tokenized into smaller units, like words or sub words, and these tokens are converted into numerical representations called embeddings

Consider the sentence: "The cat sat on the mat."

tokens: ["The", "cat", "sat", "on", "the", "mat"]

Numerical Representation "The" -> [0.2, 0.4, -0.1]

- ❖ Positional encodings are introduced. They are a set of vectors, usually generated using sinusoidal functions, which carry information about the position of each word in the sequence.

Position 1: [0.2, 0.3, 0.1]

- ❖ These positional encodings are element-wise added to the word embeddings for each token in the sentence. The result is a combined representation that includes both the meaning of the words and information about their positions.

Word Embedding for "The" $\rightarrow [0.2, 0.4, -0.1]$

Positional Encoding for Position 1 $\rightarrow [0.2, 0.3, 0.1]$

Combined Representation $\rightarrow [0.4, 0.7, 0.0]$

- ❖ By adding positional encodings to word embeddings, the model can distinguish between words based on their positions, ensuring that it understands both the meaning of the words and their order within the sentence. This allows the model to differentiate between tokens based on their position within the input.
- ❖ Parallel Attention Heads: The enable the model to focus on different parts of the input simultaneously.

- ❖ **Learning Different Patterns:** Each attention head is capable of learning different patterns and dependencies within the data
- ❖ Each attention head operates independently of the others. This means they have their own sets of learned weights and parameters. As a result, they can specialize in identifying specific types of patterns or dependencies within the input sequence.
- ❖ **Concatenation or Combination:** results from all the heads are typically concatenated or linearly combined to produce a richer representation of the input.
- ❖ **Context Management:** Within the NLP module, this component is responsible for preserving a continuous conversation context, ensuring that the chatbot maintains a coherent understanding and delivers relevant responses.
- ❖ **Layer Normalization:** Layer normalization is applied to each sub-layer's output. It helps stabilize and speed up training



Testing

can you give something about mental health

Mental health is an important aspect of overall well-being. Schools can play a crucial role in promoting mental health and well-being of students ¹. Children may have trouble communicating with others both at school and at home, which may lead to poor self-esteem, poor academic and social success, and a high dropout rate ². Separation anxiety is a common issue among children, which can affect their daily activities and tasks like going to school or peer interaction. It is characterized by experiencing extreme anxiety or even having panic attacks and completely hampers the functionality of a child ². Attachment may be understood as a bond between children and their parents or caregivers that affects the child's growth and their ability to build meaningful relationships in life. Caregivers or parents may notice that a child has problems with emotional attachment as early as their first year of birth. However, with care and patience, it is possible to overcome attachment challenges.

2 references >

AI-generated content may be incorrect

can you translate the above answer in frenh language

Type a new question...





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can you translate the above answer in french language

La santé mentale est un aspect important du bien-être général. Les écoles peuvent jouer un rôle crucial dans la promotion de la santé mentale et du bien-être des élèves ¹. Les enfants peuvent avoir des difficultés à communiquer avec les autres à l'école et à la maison, ce qui peut entraîner une faible estime de soi, une faible réussite scolaire et sociale et un taux d'abandon élevé ². L'anxiété de séparation est un problème courant chez les enfants, qui peut affecter leurs activités quotidiennes et leurs tâches comme aller à l'école ou interagir avec leurs pairs. Elle se caractérise par une anxiété extrême ou même des crises de panique et entrave complètement la fonctionnalité d'un enfant ². L'attachement peut être compris comme un lien entre les enfants et leurs parents ou soignants qui affecte la croissance de l'enfant et sa capacité à construire des relations significatives dans la vie. Les soignants ou les parents peuvent remarquer qu'un enfant a des problèmes d'attachement émotionnel dès sa première année de vie. Cependant, avec soin et patience, il est possible de surmonter les défis de l'attachement ².

2 references >

AI-generated content may be incorrect



can you give ways to improve mental state and be stress-free in tabular form with columns as [point number,name,explanation,example]

Point Number	Name	Explanation	Example
1	Exercise	Exercise is a great way to improve your mental state and reduce stress. It releases endorphins, which are natural mood boosters.	Going for a run, doing yoga, or taking a dance class.
2	Mindfulness	Mindfulness is the practice of being present in the moment and focusing on your thoughts and feelings without judgment. It can help reduce stress and anxiety.	Meditation, deep breathing exercises, or yoga.
3	Social Support	Spending time with friends and family can help improve your mood and reduce stress.	Going out for dinner with friends, calling a family member, or joining a social club. ¹
4	Self-Care	Taking care of yourself is important for your mental health. This includes getting enough sleep, eating	Taking a relaxing bath, reading a book, or getting a massage. ¹

Type a new question...





can you give it in points as heading and explanation for each in concise

Sure, here are some strategies to overcome anxiety, isolation, depression, and fear:

1. Seek professional help from a therapist or counselor .
2. Practice mindfulness and meditation .
3. Engage in regular exercise ¹ .
4. Get enough sleep ¹ .
5. Eat a healthy diet .
6. Connect with others through social activities or support groups ¹ .
7. Identify and challenge negative thoughts and beliefs .
8. Set achievable goals ¹ .
9. Practice self-care and self-compassion .

It is important to remember that everyone's experience is unique, and what works for one person may not work for another. If you are struggling with mental health concerns, it is important to seek help from a qualified professional ¹ .

1 reference >

AI-generated content may be incorrect



Type a new question...



Chatbot Metrics

- ❖ **Response Accuracy:** This measures how often the chatbot provides correct and relevant answers to user queries. You can evaluate accuracy by comparing chatbot responses to a set of expected or correct answers.
- ❖ **Response Time:** Response time is the time it takes for the chatbot to provide an answer after a user query. Faster response times are generally preferred for a better user experience.
- ❖ **Conversation Length:** Monitor the number of interactions required to resolve a user query. A shorter conversation length typically signifies more efficient communication.
- ❖ **Engagement Metrics:** These include the average session duration, the number of interactions per session, and the frequency of return visits. High engagement suggests that users find the chatbot valuable and are returning for further interactions.
- ❖ **Error Rate:** Measure the rate at which the chatbot generates incorrect or nonsensical responses. Lower error rates are desirable for maintaining the chatbot's reliability

Conclusion and Future Applications

In conclusion, our project represents a significant step forward in the field of student counseling for mental health. The synergistic integration of AI-driven chatbots, Azure services, and a tailored AI model has enabled us to provide rapid, contextually enriched responses, effectively addressing the immediate mental health concerns of students. Looking ahead, the potential applications of this technology are expansive and extend well beyond educational institutions. Its adaptability and scalability make it an ideal candidate for a broader spectrum of implementations, including telehealth services, employee assistance programs, community support initiatives, and personalized well-being applications. As we continue to refine and expand the capabilities of this system, we foresee its transformative impact on the landscape of mental health support, breaking down traditional barriers and setting new standards for accessible and responsive care.